GRADE 10 **WEDNESDAY, 14 AUGUST 2024**

MATHEMATICS TEST

TIME: 70 min **TOTAL: 75 MARKS**

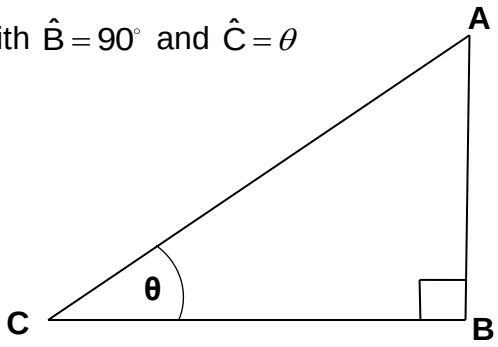
NAME: _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of 9 pages and 11 questions.
2. Answer all the questions on the question paper.
3. **All the necessary working details must be clearly shown.**
4. Approved non-programmable calculators may be used except where otherwise stated.
5. It is in your own interest to write legibly and to present your work neatly.
6. Round off answers correct to **2 decimal places** where necessary.

QUESTION 1

Study the diagram alongside. Given ΔABC with $\hat{B} = 90^\circ$ and $\hat{C} = \theta$



Complete the following:

(a) $\sin\theta = \frac{AB}{?}$

AC (1)

(b) $\frac{AB}{BC} = ?$

$\tan\theta$ (1)

(c) $\cos(90° - \theta) = \frac{?}{AC}$

AB (2)

(d) If $AB = 10cm$ and $\theta = 30^\circ$, determine the value of BC. (2)

$\tan 30^\circ = \frac{10}{BC}$

$BC = 10\sqrt{3}$

[6]

QUESTION 2

(a) In which quadrant does θ lie if:

(1) $\cos\theta < 0$ and $180^\circ < \theta < 360^\circ$? Q3 (1)

(2) $\sin\theta > 0$ and $\tan\theta < 0$? Q2 (1)

(b) If $A = 25{,}34^\circ$ and $B = 134{,}66^\circ$, determine $\cos^2(A + B)$ to two decimal places. (2)

$\cos^2(24{,}34^\circ + 134{,}66^\circ)$

$= 0{,}88$

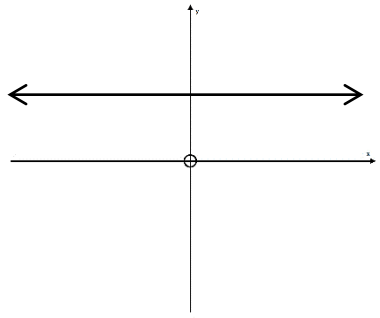
[4]

QUESTION 3

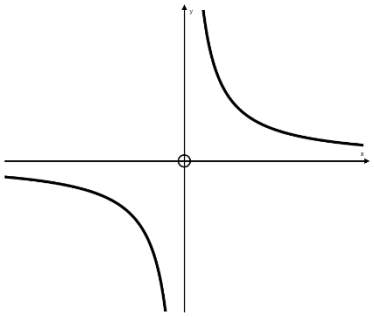
Consider each graph sketched below and match with the appropriate equation chosen from the list.

a) $y = 4 + x$ b) $y = 2$ c) $xy = 2$ d) $y = 3^x - 1$

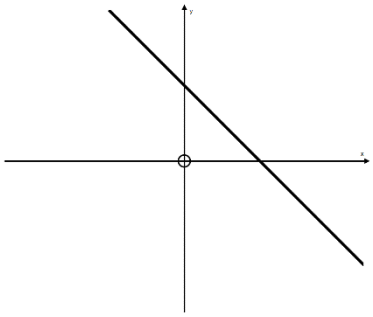
e) $y = -x^2 + 1$ f) $y = 4 - x$ g) $y = x^2 - 1$ h) $x = 2$



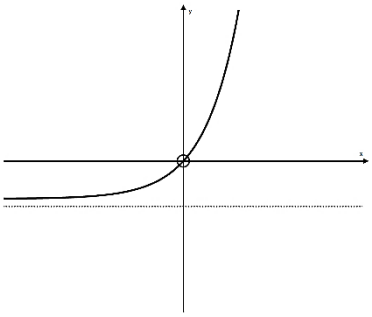
1. _____$y = 2$_____



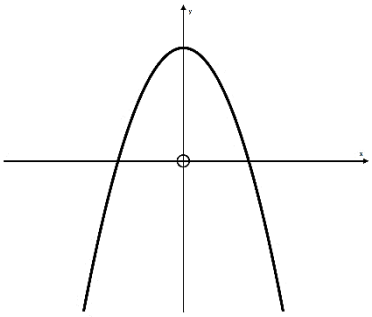
2. ____$xy = 2$ _____



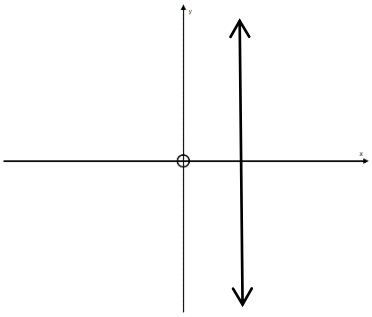
3. ____$y = 4 - x$ _____



4. _____$y = 3^x - 1$_____



5.____ $y = -x^2 + 1$_____



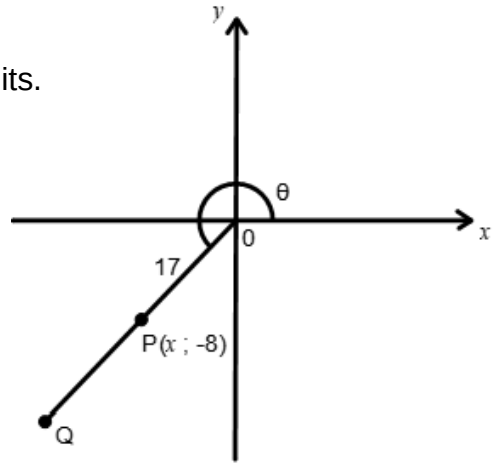
6._____ $x = 2$_____

[6]

QUESTION 4

In the diagram, P is the point (x ; -8) and OP = PQ = 17 units.

Determine, without the use of a calculator:



(a) the value of x. (3)

$(17)^2-(-8)^2=x^2$ (Pythagoras)

$x=15$

$\therefore x=-15$

(b) $\cos\theta.\tan\theta$ (3)

$\left(\frac{-15}{17}\right)\left(\frac{-8}{-15}\right)$

$=-\frac{8}{17}$

(c) the co-ordinates of Q. (2)

$OQ=34$

$\therefore Q(-30\ ;\ -16)$

[8]

QUESTION 5

Solve for β if $\beta\in[0°;90°]$.

Answer correct to 2 decimal places.

(a) $\sin\beta=0,604$ (1)

$\beta=\sin^{-1}(0,604)$

$\beta=37,16°$

(b) $\sin(\beta+40°)=0,86$ (2)

$\beta+40°=\sin^{-1}(0,86)$

$\beta+40°=59,31658...°$

$\beta=19,32°$

(c) $4\tan\beta=7-4\tan\beta$ (3)

$8\tan\beta=7$

$\tan\beta=\frac{7}{8}$

$\beta=\tan^{-1}\left(\frac{7}{8}\right)$

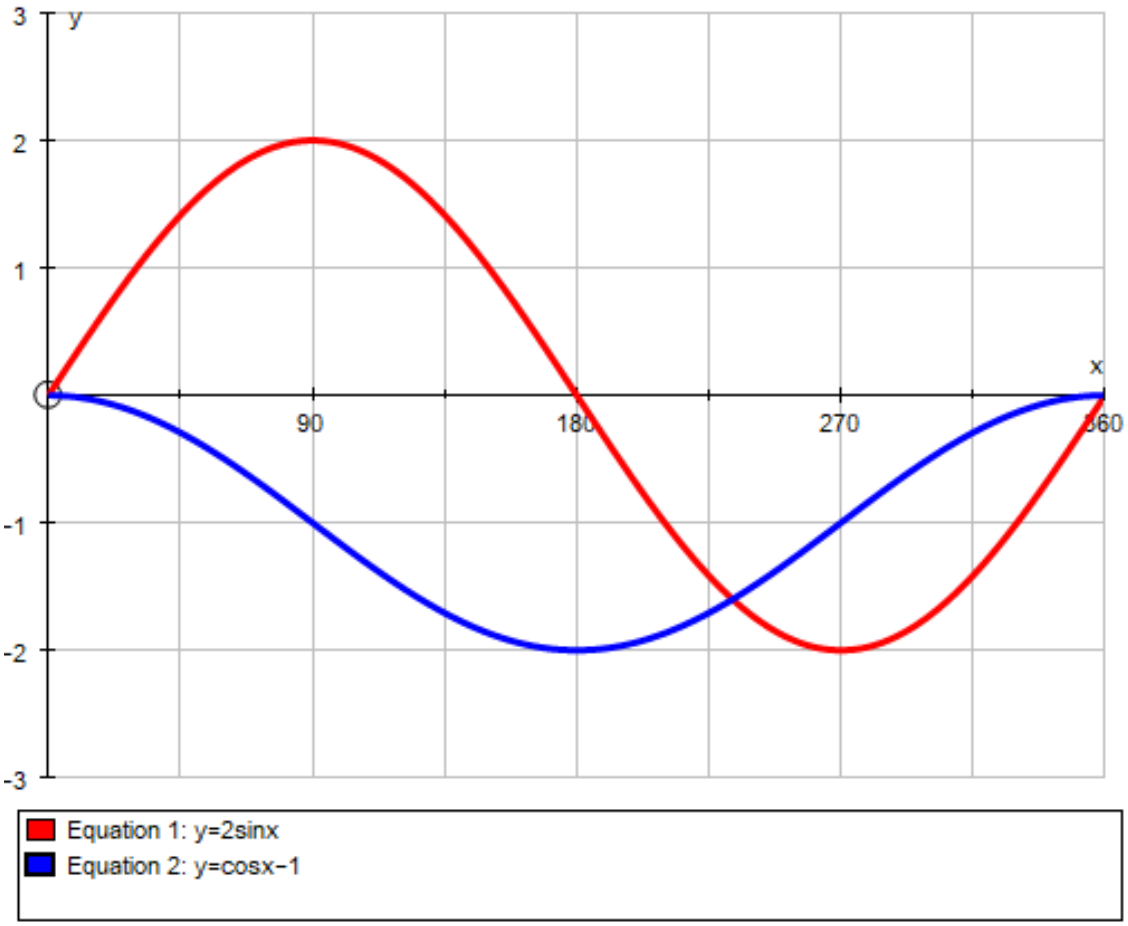
$\beta=41,19°$

[6]

QUESTION 6

Given: $f(x) = 2\sin x$ and $g(x) = \cos x - 1$ where $x \in [0°;360°]$

(a) Sketch the graphs of f and g on the axes below. Label each graph clearly. (6)



(b) Determine:

(1) the period of $f(x)$. (1)

$360°$

(2) the minimum of $g(x)$. (1)

-2

(c) If the graph of $g(x)$ is translated (shifted) 3 units down, what would the equation of the new graph be? (2)

$g'(x) = \cos x - 4$

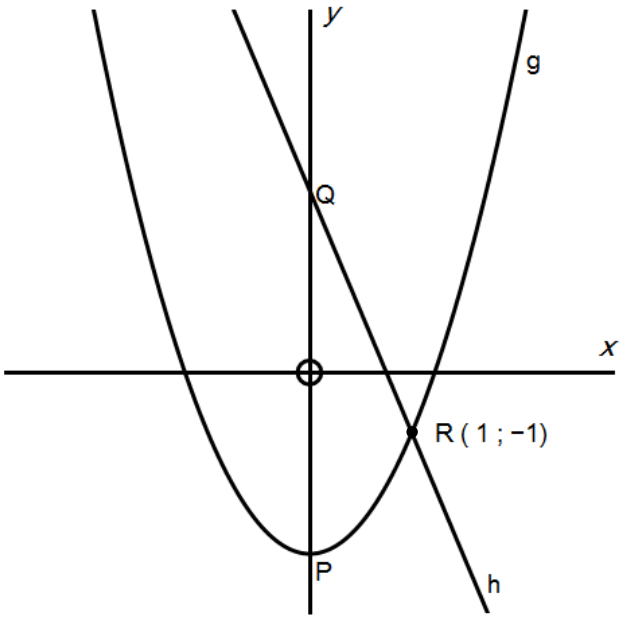
[10]

QUESTION 7

Sketched are $g(x)=ax^2+q$ and $h(x)=mx+c$.

Q and $P(0;-3)$ are the *y* intercepts of *h* and *g* respectively.

$R(1;-1)$ is the point of intersection of *g* and *h*.



(a) Determine the equation of *g*. (3)

$g(x)=ax^2-3$

Sub $R(1;-1)$

$-1=a(1)^2-3$

$a=2$

$\therefore g(x)=2x^2-3$

(b) Write down the range of *g*. (2)

$y\in[-3,\infty)$

(c) For which values of *x* is *g* increasing? (2)

$x\in(0;\infty)$

(d) It is further given that $PQ=6$ units, determine the equation of *h*. (3)

$h(x)=mx+3$

Sub $R(1;-1)$ ***OR gradient formula

$-1=1m+3$

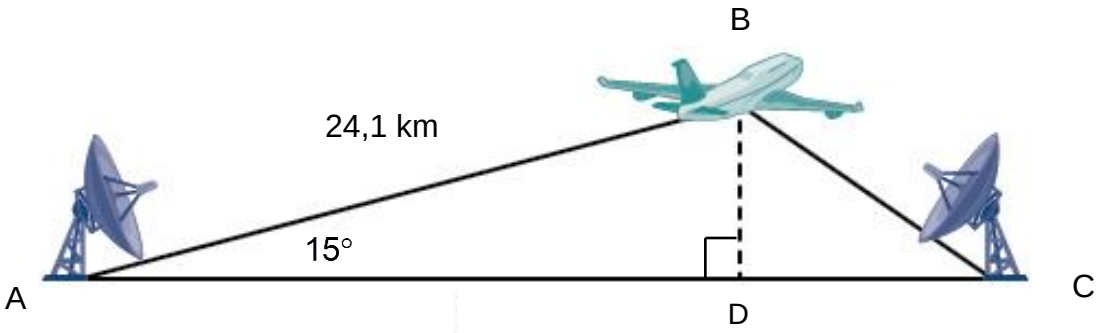
$m=-4$

$h(x)=-4x+3$

[10]

QUESTION 8

The given diagram is a representation of a passenger aircraft shortly after take-off.
$AC = 32,2$ km ; $A\hat{D}B = 90°$; $B\hat{A}D = 15°$ and $AB = 24,1$ km.



Calculate, correct to one decimal place:

(a) BD, the height of the plane above the ground. (3)

$\sin 15^\circ = \dfrac{BD}{24,1}$

$BD = 6,2km$

(b) $D\hat{C}B$ (5)

For AD:

$\cos 15^\circ = \dfrac{AD}{24,1}$

$AD = 23,3km$ (can use Pyth)

$\therefore CD = 32,2 - 23,3$

$CD = 8,9$

$\therefore In\ \Delta BDC$

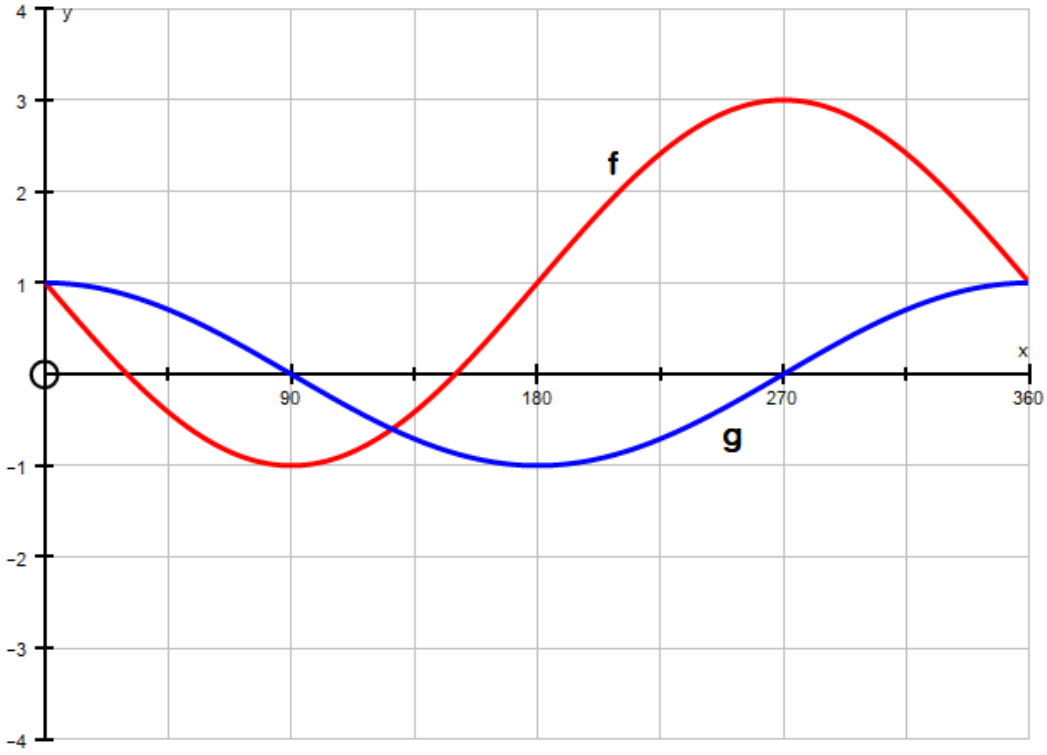
$\tan \hat{C} = \dfrac{6,2}{8,9}$

$\hat{C} = 34,9^\circ$

[8]

QUESTION 9

The diagram below represents the functions of $f(x) = a\sin x + c$ and $g(x) = d\cos x$ for $x \in [0^\circ;360^\circ]$



(a) Write down the values of a,c and d. (3)

$a = -2$

$c = 1$

$d = 1$

(b) What is the amplitude of f? (1)

2

(c) Give two values of x where $f(x) = g(x)$. (2)

$x = 0^\circ$

$x = 360^\circ$

$x = \pm 130^\circ$

(d) For what value of x is $g(x) - f(x) = 1$? (1)

$x = 90^\circ$

(e) For what value(s) of x is $g(x) \geq 0$ if $x \in [180^\circ;360^\circ]$. (2)

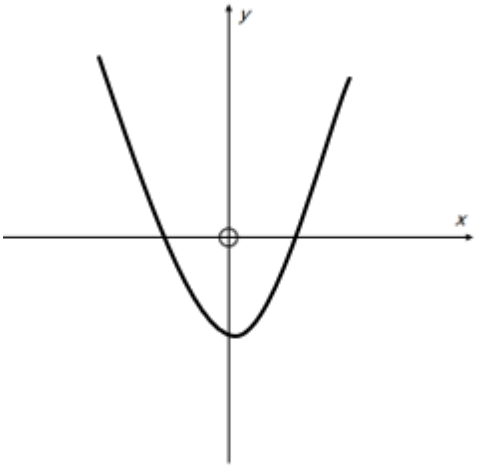
$x \in [270^\circ,360^\circ]$

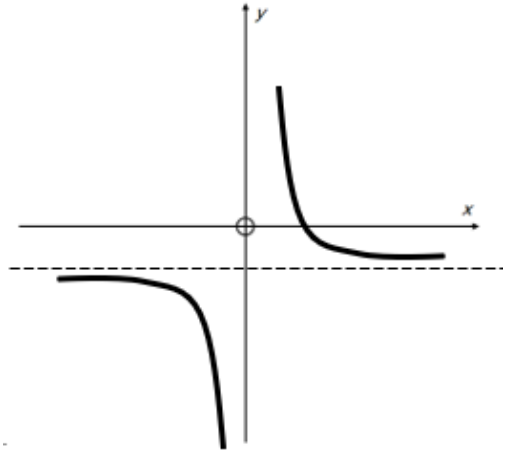
[9]

QUESTION 10

Draw a rough sketch of the following, each on a separate axis:

(a) $y = ax^2 + q$ where $q < 0$ and $a > 0$ (b) $y = \frac{a}{x} + q$ where $a > 0$ and $q < 0$





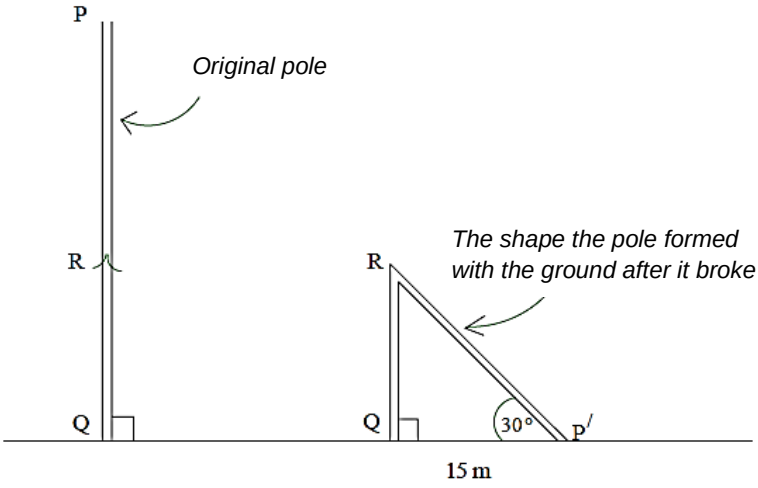
[4]

QUESTION 11

A pole, PQ, broke at point R which resulted in the top portion of the pole, PR, forming an angle of $30°$ with the ground at P', now 15 m away from Q, the foot of the pole.

Calculate PQ, the original height of the pole.

Write your answer in surd form.



For RQ:

$\tan 30^\circ = \frac{RQ}{15}$

$RQ = 5\sqrt{3}$

For RP':

$RP'^2 = RQ^2 + QP'^2$ (Pythagoras) ***Can use $\cos 30^\circ = \frac{15}{RP'}$

$RP' = 10\sqrt{3}$

$\therefore PQ = 10\sqrt{3} + 5\sqrt{3}$

$PQ = 15\sqrt{3}$

[4]